Class-Aware Drift Compensation for Non-Uniform Semantic Shift in Continual Learning

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Abstract

Continual learning aims to develop models that can incrementally acquire new knowledge without sacrificing previously acquired information. When the storage of past data is completely prohibited, catastrophic forgetting is exacerbated, stimulating research in exemplar-free continual learning, where class prototypes are typically retained and periodically updated via drift compensation. However, these drift compensation methods generally assume that semantic shift between tasks occurs uniformly (i.e., uniformly mild or uniformly abrupt), whereas in real-world data streams, such shift is often unpredictable. To address this issue, we propose a Class-Aware Drift Compensation (CADC) method to handle feature drift under non-uniform semantic shift. The method dynamically models the drift degree of old classes under the challenging exemplar-free scenario, enabling class-specific adaptive compensation. Furthermore, we demonstrate that the class drift modeled by CADC can effectively guide knowledge distillation by imposing class-specific constraints on logits based on drift intensity, which motivates the design of drift-guided distillation to better balance model stability and plasticity. Extensive experiments on six datasets demonstrate the superiority of the proposed method over existing exemplar-free approaches and the advantage of CADC over state-of-the-art drift compensation techniques. Code is available at https://github.com/AldrinLake/CADC.git.

1. Introduction

Neural network models have demonstrated outstanding performance in a wide array of real-world applications, including computer vision, natural language processing, and multimodal learning [14, 27, 34]. In practice, as operational data gradually accumulates in evolving systems, models are often required to continuously acquire new patterns from incremental inputs. However, the dynamic and non-i.i.d. nature of such data often leads to suboptimal learning, as gradient updates from new data may interfere with previously learned knowledge [40]. This phenomenon, known as the notorious *catastrophic forgetting*, has attracted significant attention in the field of continual learning [2, 16, 36, 48], which aims to develop models that strike a balance between stability and plasticity by acquiring new knowledge without sacrificing previously learned information.

As a representative paradigm in continual learning, Class Incremental Learning (CIL) [5, 10, 19] defines a unified

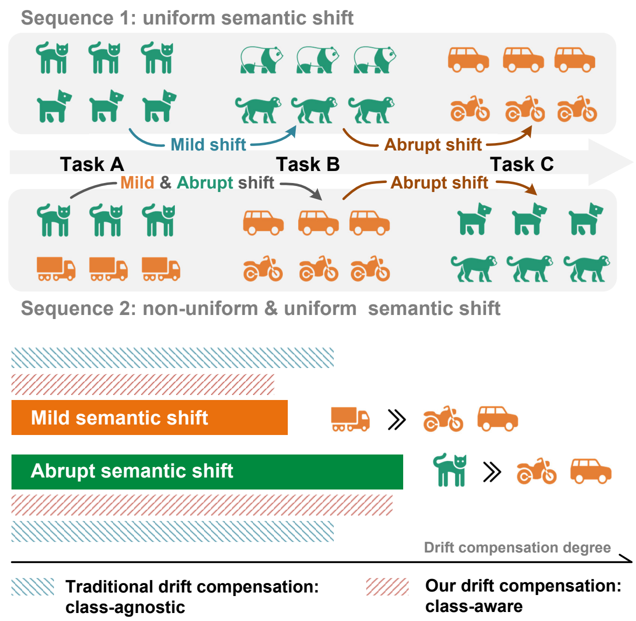

Figure 1. A motivational illustration of class-aware drift compensation. Objects of the same color share semantic similarity, such as *cats* and *pandas* as animals, and *trucks* and *sedans* as vehicles. The top part shows that classes across tasks may be semantically related or unrelated, indicating non-uniform semantic shift. The bottom part compares the semantic shift degrees across different classes in Task A during the transition to Task B in Sequence 2.

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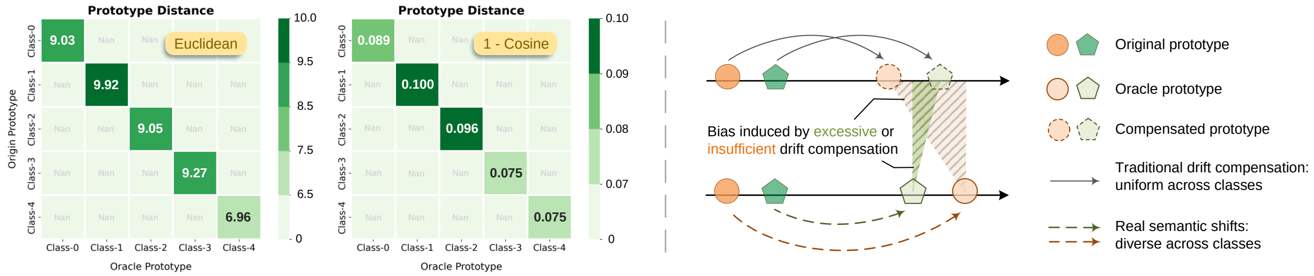

Figure 2. At the left, heatmaps illustrate the degree of semantic shift. Experiments are conducted with LwF [23] on CIFAR-100 (initial 0 classes, increments of 5 classes). Oracle prototypes are recalculated using data from the first task upon completion of the second task, whereas the original prototypes are computed immediately after the first task is finished. Significant variations in semantic drift can be observed among different classes based on both Euclidean and cosine distances. The right illustrates that applying uniform drift compensation across classes leads to excessive or insufficient correction.

framework in which a model is trained sequentially on a series of tasks, each comprising mutually exclusive classes, while being expected to perform well across all previously encountered classes at inference [20, 21, 39]. A common strategy in CIL is to store a limited number of exemplars per class in a memory buffer for replay during subsequent tasks. Although effective, this approach faces constraints in storage and data privacy, driving growing interest in exemplar-free solutions [35, 47]. In this work, we focus on Exemplar-Free Class Incremental Learning (EFCIL) under a cold-start setting, which poses a significant challenge as the model is trained from scratch without access to any prior knowledge. In EFCIL, prototypes or intermediate features are often stored to support the model's ability to learn continuously [15, 22, 29].

However, a key challenge in EFCIL is that the model's ongoing evolution causes stored representations to become outdated, leading to semantic feature drift [7, 37]. Addressing such drift remains a critical challenge, and recent studies have strived to alleviate forgetting by correcting old class representations through drift compensation [9, 13, 31]. Although effective, these methods assume uniform semantic shifts between tasks, meaning the shifts are either uniformly mild or uniformly abrupt [15]. However, semantic relationships between classes across tasks are typically random, resulting in largely unpredictable semantic shifts for past classes during task transitions. The phenomenon can be illustrated in Fig. 1. In Sequence 1, both Task A and Task B involve animal classes, leading to uniformly mild semantic shifts in Task A. Similarly, all classes in Task B undergo uniformly abrupt shifts when transitioning to Task C. Such uniform transitions represent the ideal case assumed by existing drift compensation approaches. However, these methods are less effective in the more realistic scenario of Sequence 2, where the transition from Task A to B results in non-uniform shifts. Since Task B focuses on vehicles, the class *cat* from Task A experiences an abrupt semantic shift, whereas the class *truck* undergoes only a mild one. The presence of such non-uniform semantic shifts is verified by measuring the distance between stored and oracle prototypes on CIFAR-100, as shown in Fig. 2. As unified drift compensation is class-agnostic, it may lead to excessive or insufficient adjustments under non-uniform semantic shifts, with bias accumulating over tasks.

To address the above challenge, we propose a Class-Aware Drift Compensation (CADC) method tailored to complex non-uniform semantic shifts. It adaptively corrects the intermediate representations of different classes based on semantic shift degrees. Since CADC relies on estimating class-specific semantic shifts and past data are unavailable in the exemplar-free scenario, it cannot evaluate semantic shifts using oracle prototypes in the same way as the heatmaps shown in Fig. 2. We propose dynamically learning drift compensation and treating the compensated old class representations in each training round as approximations of oracle prototypes, thereby enabling real-time evaluation of drift. In this process, the updated prototypes guide the modeling of semantic shift degrees, which in turn direct the compensation of prototypes. Although this forms a seemingly suboptimal loop, prototype gradients are detached via the drift compensation network, and the model still shows strong empirical performance. Moreover, we show that the class-specific semantic shift can inform knowledge distillation by applying tailored constraints to logits according to the drift degrees, leading to the design of a drift-guided knowledge distillation. The key contributions are summarized as follows:

1. We propose CADC, the first drift compensation method designed to address non-uniform semantic shifts across tasks, enabling precise correction for each class.
2. A drift-guided knowledge distillation is developed to dynamically adjust class-wise regularization strength according to different semantic shifts.
3. Extensive experiments show that CADC outperforms existing EFCIL methods and achieves superior results compared to existing drift compensation approaches.

2. Related Work

Class Incremental Learning (CIL). In CIL, a model is trained sequentially on a series of tasks with disjoint classes. Existing approaches to CIL can be broadly categorized into three types [42, 43]: regularization-based methods, dynamic network-based methods, and rehearsal-based methods. The method based on regularization aims to preserve prior knowledge by explicitly or implicitly constraining changes in model parameters [1, 26]. Dynamic network-based methods adopt the principle of parameter isolation by expanding the architecture for each task [4, 6]. The method based on rehearsal alleviates forgetting by introducing buffered samples from previous tasks during new task training [44]. Despite the effectiveness, the rehearsal-based method is limited by memory constraint or privacy concerns. As a result, exemplar-free class incremental learning (EFCIL) has emerged as an increasingly prominent research focus [12, 38]. Since past data are strictly prohibited from being retained, the distributions of previous classes in the new feature space remain unknown, presenting significant challenges related to feature drift. In this work, we adopt a cold-start setting, enabling a more direct evaluation of the drift compensation performance.

Drift Compensation. As the feature extractor is continuously updated in CIL, previously stored representations become invalid in the new feature space, resulting in feature drift. Drift compensation [13] provide a practical solution to the feature drift issue by estimating the distribution of previous classes within the new feature space. SDC [37] suggests that the drift of old class representations can be estimated by analyzing distributional shifts of new task data across the old and new feature spaces. This finding has spurred increasing attention to drift compensation [22, 30]. Recently, LDC [7] learns a projector between the outputs of the current and previous models, and projects old class prototypes into the new representation space after each task. However, existing methods assume uniform semantic shifts, overlooking that different classes may shift differently. Uniform compensation often leads to excessive or insufficient correction, with such bias accumulating over time. To our knowledge, this is the first work to address drift compensation under the non-uniform semantic shift.

3. Methodology

3.1. Preliminaries

Notation. In CIL, the model learns a sequence of tasks $\mathcal{T} = \{T_1, T_2, \ldots, T_N\}$, where N denotes the number of tasks. The task $T_t \in \mathcal{T}$ consists of a training set $\mathcal{D}_t^{train} = \{(x_i, y_i)\}_{i=1}^{n}$ associated with a set of classes $\mathcal{C}_t = \{c_1, c_2, \ldots, c_m\}$, where $y_i \in \mathcal{C}_t$ corresponds to the ground-truth class of x_i. For $\forall T_t, T_k \in \mathcal{T}$, their label spaces satisfy $\mathcal{C}_t \cap \mathcal{C}_k = \emptyset$. The test data $\mathcal{D}_t^{test}$ for task T_t includes all classes observed from T_1 to T_t, denoted as $\mathcal{C}_{1:t}$. A deep model of task T_t can be viewed as a composition of a feature extractor f_θ^t parameterized by θ and a classifier g_π^t parameterized by π. f_θ^t learns a d dimension representation z_i of input x_i, i.e., $z_i = f_\theta^t(x_i) \in \mathbb{R}^d$. A unified classification head g_π^t maps the representation z_i to a probability distribution over all seen classes, indicated as $g_\pi^t(z_i) \in \mathbb{R}^{|\mathcal{C}_{1:t}|}$. The model for task T_t is initialized from task T_{t-1}.

Problem Definition. This work focuses on addressing the feature drift problem [9] in CIL, where continuous model updates cause semantic shifts in previously learned representations. This issue is more severe in exemplar-free settings, where no stored samples anchor the old feature space. Drift compensation [37] mitigates the feature drift issue by aligning old class representations with the evolving feature space, thereby preserving their semantic consistency during continual learning. This work reveals that the semantic shifts of old-class representations are non-uniform [15] across different classes, and drift compensation should be performed in a class-aware manner according to the extent of feature drift. The vanilla drift compensation approach learns a linear mapping from the old representation space to the new one after the new model has been fully adapted to the new task. Its objective for adapting to the new task T_t is

$$\mathcal{O}_{\text{adapt}} = \inf_{\theta,\pi} \mathbb{E}_{x\sim\mathcal{D}_t^{train}} \left[\mathcal{L}_{\text{cls}} + \gamma \cdot \mathcal{L}_{\text{kd}}\right] \tag{1}$$

where $\mathcal{L}_{\text{cls}}$ is the cross-entropy loss for learning new classes, and $\mathcal{L}_{\text{kd}}$ is the logits-based distillation loss for preserving old knowledge [23]. γ is the balancing parameter. In particular, we introduce $\mathcal{L}_{\text{trsf}}$ in $\mathcal{O}_{\text{adapt}}$ to dynamically model the degree of semantic shift, which guides both drift compensation and knowledge distillation.

3.2. Motivation

Existing drift compensation methods typically rely on the assumption of uniform semantic shifts across tasks [15]. In practice, semantic shift varies across classes, causing vanilla methods to excessively or insufficiently compensate, with bias accumulating over time. This observation motivates the design of our Class-Aware Drift Compensation (CADC) approach, as illustrated in Fig. 3. Fig. 3 illustrates the training process for a new task T_t. A default cross-entropy loss is applied for learning new classes (omitted for clarity), alongside two key modules: drift-guided knowledge distillation (KD) and the drift estimator. The drift estimator, implemented as a linear layer, models the transfer pattern in the representation space by fitting outputs from the old and new models. This pattern is then used to perform drift compensation, continuously updating old prototypes during training. By computing discrepancies between prototypes in the old and new representation spaces, we dynamically model semantic shifts and obtain the class-wise

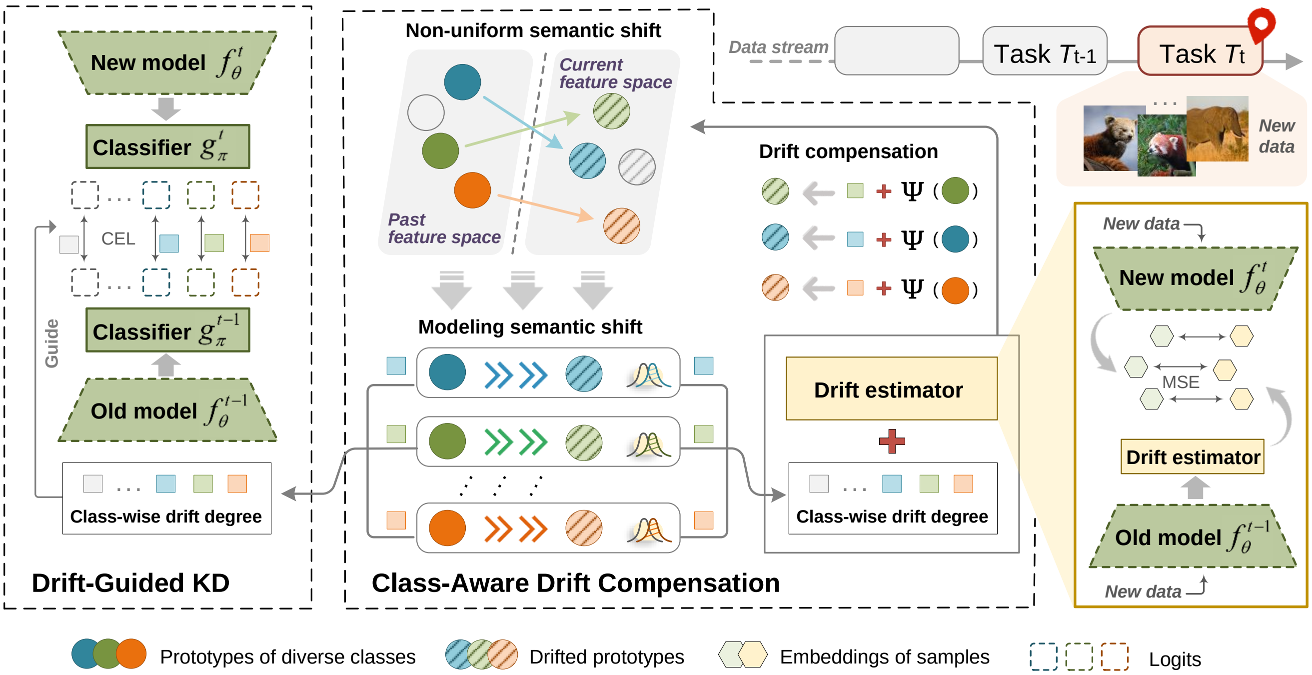

Figure 3. Overview of the proposed method. Class-aware drift compensation is achieved by modeling non-uniform semantic shifts via prototype differences between old and new feature spaces, yielding class-wise drift degrees. These degrees facilitate prototype alignment to the new feature space through drift compensation. Additionally, class-wise drift degrees guide knowledge distillation by assigning stronger regularization to classes with higher transfer difficulty.

drift degree, which guides both CADC and KD. As gradients to old prototypes are detached when passing through the drift estimator, this design avoids a non-convergent loop and achieves empirically strong performance.

3.3. Class-Aware Drift Compensation

As previously mentioned, semantic shifts between tasks vary across classes, so an ideal drift compensation method should tailor the correction according to the class-specific semantic shifts. Thus, modeling semantic shift is essential for the success of class-aware drift compensation.

Modeling Semantic Shift. Let $\mathbb{P}^i$ denote the original probability distribution of old class $c_i \in \mathcal{C}_{1:t-1}$, and $\mathbb{Q}^i$ its distribution in the feature space of the current task T_t. The semantic shift of an arbitrary class $c_i \in \mathcal{C}_{1:t-1}$ can be reflected by modeling the optimization objective that preserves its knowledge, expressed as

$$\mathcal{O}_{\text{shift}}^{c_i} = \inf_{\theta} \int_{\mathcal{X}^{c_i}} \mathbb{P}^i \left(f_\theta^{(c_i)}(x) \right) \log \frac{\mathbb{P}^i \left(f_\theta^{(c_i)}(x) \right)}{\mathbb{Q}^i \left(f_\theta^t(x) \right)}, \quad (2)$$

where (c_i) denote the task index to which class c_i belongs, and $\mathcal{X}^{c_i}$ represent the input space of c_i. Ideally, minimizing the semantic shift of class c_i in Eq. (2) maximizes the preservation of knowledge for c_i. The feature drift caused by semantic shift is a major challenge in CIL. Although freezing the feature extractor, i.e., setting $f_\theta^t = f_\theta^{(c_i)}$, can avoid this issue, it limits the model's plasticity for learning new classes. Therefore, permitting a controlled degree of drift while implementing precise compensation for each class is a viable strategy. However, in EFCIL, the use of past data is prohibited, making it infeasible to directly compute the true semantic shift degree $\mathcal{S}^t(c_i)$ for $c_i \in \mathcal{C}_{1:t-1}$. An alternative is to estimate it by measuring the distance between class prototypes in the old and new feature spaces. However, the oracle prototypes in the new space are still inaccessible without old data. To address this, we propose approximating the position of old class prototypes in the new representation space gradually using a drift estimator Ψ. Let $\mathrm{p}^*_i \in \mathrm{P}^*_{1:t-1} = \{\mathrm{p}^*_i\}_{i=1}^{|\mathcal{C}_{1:t-1}|}$ denote the oracle prototype of class $c_i \in \mathcal{C}_{1:t-1}$ in task T_t, and $\mathrm{p}_i \in \mathrm{P}_{1:t-1} = \{\mathrm{p}_i\}_{i=1}^{|\mathcal{C}_{1:t-1}|}$ its original prototype. Thus, the semantic shift for class c_i can be computed as

$$\mathcal{S}^t(c_i) = \| \mathrm{p}_i - \mathrm{p}^*_i \|_2 = \| \mathrm{p}_i - \Psi(\mathrm{p}_i) \|_2, \quad (3)$$

where $\|\cdot\|_2$ denotes the l_2 norm, i.e., the Euclidean distance. Other metrics such as cosine distance can also be considered, which will be discussed in the experimental section. The drift estimator Ψ, realized as a linear layer $\mathbf{W} \in \mathbb{R}^{d\times d}$ with bias $\mathbf{b} \in \mathbb{R}^d$, is endowed with the ability to capture the transformation pattern between old and new feature spaces through the following optimization objective

$$\mathcal{O}_{\mathrm{trsf}} = \inf_{\mathbf{W},\mathbf{b}} \mathbb{E}_{x\sim\mathcal{D}_t^{train}}[\mathcal{L}_{\mathrm{trsf}}(x; f_\theta^t, f_\theta^{t-1}, \mathbf{W}, \mathbf{b})], \quad (4)$$

$$\mathcal{L}_{\mathrm{trsf}} = \frac{1}{B}\sum_{i=1}^{B}\left(\left(f_\theta^{t-1}(x_i)\cdot\mathbf{W}+\mathbf{b}\right) - f_\theta^t(x_i)\right)^2, \quad (5)$$

where B denotes the batch size, and the loss is computed as the mean squared error (MSE). The loss $\mathcal{L}_{\mathrm{trsf}}$ in the optimization objective depends only on the current task data $\mathcal{D}_t^{train}$, and the resulting gradients affect only the parameters $\{\mathbf{W}, \mathbf{b}\}$ of the drift estimator Ψ. By learning the transfer pattern from the old to the new feature space via Eq. (4), the drift estimator is expected to align the old class prototypes with the new space, meaning $\Psi(\mathrm{p}_i) \simeq \mathrm{p}^*{}_i$ for $\forall \mathrm{p}_i \in \mathrm{P}_{1:t-1}$. The optimization objective in Eq. (4) is performed concurrently during the learning of new task T_t, allowing the semantic shift degree in Eq. (3) to be dynamically modeled at each epoch.

Drift Compensation. Before applying class-aware drift compensation to update prototypes, compensation weights $\{\tau_i\}_{i=1}^{|\mathcal{C}_{1:t-1}|}$ should be assigned to classes based on Eq. (4). In particular, the weight τ_i is adaptively computed based on the dynamically modeled semantic shift degree $\mathcal{S}^t(c_i)$. Specifically, for $\forall \tau_i \in \{\tau_i\}_{i=1}^{|\mathcal{C}_{1:t-1}|}$, it is defined as

$$\tau_i = \frac{\epsilon}{1+\exp\left(-\bar{\mathcal{S}}^t(c_i)\right)}, \ \bar{\mathcal{S}}^t(c_i) = \frac{\mathcal{S}^t(c_i)}{\max\{\mathcal{S}^t(c_i)\}_{i=1}^{|\mathcal{C}_{1:t-1}|}}. \quad (6)$$

Here ϵ denotes a scaling factor, which will be discussed in the experimental section. $\bar{\mathcal{S}}^t(c_i)$ can be interpreted as the relative strength of semantic shift. Then, after each training epoch, the old class prototypes can be corrected through class-aware drift compensation, computed as

$$\Psi(\mathrm{p}_i) = \mathrm{p}_i + \tau_i\cdot(\mathrm{p}_i\cdot\mathbf{W}+\mathbf{b}-\mathrm{p}_i), \tau_i \leftarrow \frac{\tau_i+\tau_i'}{2}, \quad (7)$$

where τ_i can be interpreted as the class-wise drift degree, and τ_i' corresponds to its value computed in the last epoch. A momentum update strategy is adopted to refine τ_i, aiming to mitigate the impact of gradient instability on modeling semantic shifts during training. It is worth noting that during training, the gradients from Eq. (7) are detached. Consequently, as training proceeds, the new representation space becomes stable, the semantic shift estimated by Eq. (3) converges, and the class-aware drift compensation conducted by Eq. (7) reaches its optimal performance.

3.4. Drift-Guided Knowledge Distillation

In the last section, the semantic shifts of old classes are dynamically modeled during training. Building on this foundation, it is valuable to further examine the rationale behind vanilla knowledge distillation (KD) in CIL. In vanilla KD, a uniform constraint is applied to the logits of all classes across all training phases, without accounting for the varying difficulty involved in transferring knowledge of different classes. We argue that knowledge that is more difficult to transfer should be assigned stronger regularization in order to maximize its retention. Conversely, knowledge that is easier to transfer requires only a weaker degree of regularization. Inspired by this, we utilize the class-wise drift degree τ_i modeled by Eq. (7) to quantify the difficulty of knowledge transfer and accordingly design a Drift-Guided KD (DGKD), the optimization objective of which is defined with a cross-entropy loss (CEL), expressed as

$$\mathcal{O}_{\mathrm{dgkd}} = \inf_{\theta,\pi} \mathbb{E}_{x\sim\mathcal{D}_t^{train}}[\mathcal{L}_{\mathrm{dgkd}}(x; f_\theta^t, g_\pi^t)], \quad (8)$$

$$\mathcal{L}_{\mathrm{dgkd}} = -\frac{1}{B}\sum_{i=1}^{B}\sum_{j=1}^{|\mathcal{C}_{1:t-1}|}\tau_j\cdot\left(l_i^j \log \bar{l}_i^j\right), \quad (9)$$

where l_i^j and $\bar{l}_i^j$ denote the j-th logits of $g_\pi^t(f_\theta^t(x_i))$ and $g_\pi^{t-1}\left(f_\theta^{t-1}(x_i)\right)$, respectively. The drift degree τ_j evolves dynamically during training, allowing the distillation to adapt to individual classes. This adaptive mechanism promotes class-specific knowledge retention, thereby mitigating forgetting in a more precise manner.

3.5. Overall Optimization

The overall loss for the optimization of our proposed method can be summarized as follows:

$$\mathcal{L} = \mathcal{L}_{\mathrm{cls}} + \gamma\cdot\mathcal{L}_{\mathrm{dgkd}} + \delta\cdot\mathcal{L}_{\mathrm{trsf}}, \quad (10)$$

where γ and δ are hyperparameters used to balance the importance of different modules. $\mathcal{L}_{\mathrm{cls}}$ serves as the base module, employing a cross-entropy loss to learn new class knowledge. During inference, we use the prototypes updated by Eq. (7) and adopt the nearest class mean (NCM) classifier, as commonly done in prior work [7]. Its sensitivity to prototype quality makes it particularly suitable for reliably evaluating the effectiveness of drift compensation.

4. Experiments

4.1. Experiment Settings

Datasets. We conduct experiments on standard CIL benchmarks, including CIFAR-100 [18], TinyImageNet [33], and ImageNet-Subset [3], as well as two fine-grained datasets, CUB-200 [32] and Stanford Cars [17], and a large-scale dataset, ImageNet-1K [3]. Following existing EFCIL studies [7, 9], we adopt the cold start setting, where each dataset is evenly split into several tasks. Specifically, under the 5/10/20 task splits, CIFAR-100 and ImageNet-Subset contain 20/10/5 classes per task, while TinyImageNet contains

Table 1. Comparison of EFCIL methods on three datasets under 5-task and 10-task splits. Results marked with † are reported from [9].

	Methods	CIFAR-100		TinyImageNet		ImageNet-Subset	
		5 stages	10 stages	5 stages	10 stages	5 stages	10 stages
Last Task Accuracy	LwF†	45.35	26.14	38.81	27.42	50.88	37.90
	PASS	48.15 ± 0.88	25.33 ± 4.51	34.78 ± 0.39	25.07 ± 0.28	39.27 ± 0.05	26.70 ± 0.22
	SSRE	37.30 ± 1.31	18.64 ± 1.43	28.75 ± 0.10	20.78 ± 1.01	35.16 ± 0.38	23.60 ± 0.28
	FeTrIL	41.96 ± 0.34	32.38 ± 0.70	21.50 ± 0.77	15.42 ± 0.64	41.86 ± 1.27	30.45 ± 1.14
	FeCAM†	47.28	33.82	25.62	23.21	54.18	42.68
	NAPA-VQ	46.68 ± 0.34	37.07 ± 012	26.67 ± 0.15	20.37 ± 0.06	38.93 ± 0.67	26.15 ± 0.75
	FKSR	44.50 ± 0.69	31.25 ± 2.01	32.29 ± 3.98	19.03 ± 0.24	47.73 ± 0.87	35.65 ± 0.38
	FCS	45.47 ± 3.00	32.95 ± 5.08	40.85 ± 0.40	30.71 ± 2.85	58.69 ± 0.39	40.39 ± 1.70
	EFC	53.13 ± 0.59	44.65 ± 0.46	45.43 ± 0.34	34.10 ± 0.77	58.89 ± 1.17	47.38 ± 1.43
	Ours	**59.41 ± 0.35**	**47.32 ± 0.56**	**47.19 ± 0.42**	**38.88 ± 0.55**	**68.30 ± 0.90**	**57.07 ± 0.70**
Average Incr. Accuracy	LwF†	61.94	46.14	49.70	38.77	69.11	61.60
	PASS	63.32 ± 0.72	43.14 ± 4.68	45.55 ± 0.56	36.01 ± 0.16	59.87 ± 0.26	49.03 ± 0.34
	SSRE	51.86 ± 0.31	33.15 ± 1.46	40.76 ± 1.34	33.73 ± 2.08	50.24 ± 0.74	39.30 ± 0.79
	FeTrIL	59.21 ± 0.51	50.26 ± 0.15	35.25 ± 1.79	27.91 ± 2.30	57.93 ± 1.42	48.06 ± 1.18
	FeCAM†	61.37	48.58	39.85	35.32	67.21	57.45
	NAPA-VQ	61.50 ± 0.07	51.94 ± 0.31	40.50 ± 0.53	31.72 ± 0.04	55.22 ± 0.67	42.35 ± 0.84
	FKSR	60.96 ± 0.10	48.43 ± 2.08	44.50 ± 3.64	32.99 ± 2.67	62.61 ± 0.38	52.47 ± 0.06
	FCS	61.93 ± 1.83	47.60 ± 6.23	54.07 ± 0.80	45.19 ± 2.67	66.94 ± 1.52	56.12 ± 1.67
	EFC	66.79 ± 1.31	59.31 ± 1.96	57.06 ± 0.67	47.95 ± 0.61	71.63 ± 1.39	59.94 ± 1.38
	Ours	**71.56 ± 0.72**	**63.18 ± 0.96**	**59.03 ± 1.63**	**52.54 ± 1.82**	**77.43 ± 0.46**	**70.35 ± 0.84**

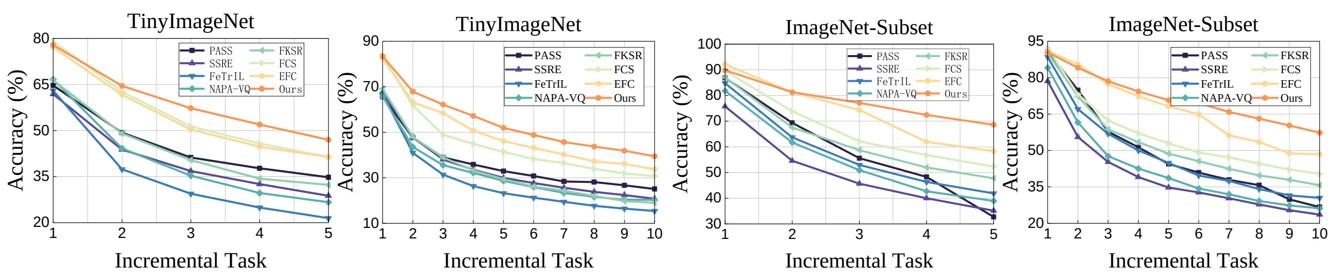

Figure 4. Test accuracy after each task for different methods on TinyImageNet, and ImageNet-Subset under 5-task and 10-task splits.

Table 2. Performance on long sequence tasks (50-task split).

Methods	TinyImageNet		ImageNet-Full	
	$\mathcal{A}_{last}$	$\mathcal{A}_{inc}$	$\mathcal{A}_{last}$	$\mathcal{A}_{inc}$
SDC	11.43	23.04	12.43	31.61
LDC	16.86	29.70	16.69	36.98
ADC	17.57	28.17	13.63	35.27
CADC	**19.46**	**30.79**	**19.09**	**39.68**

Table 3. Last task accuracy comparison on fine-grained datasets.

Methods	CUB-200		Stanford Cars	
	5 stages	10 stages	7 stages	14 stages
SDC	44.97	29.67	38.54	28.89
LDC	49.49	33.50	39.56	28.95
ADC	51.01	31.28	40.62	32.90
CADC	**51.27**	**35.36**	**41.19**	**33.51**

40/50/10 classes per task, respectively. Results for 20 tasks are reported in the Appendix.

Comparison Methods. The proposed method is compared with several existing EFCIL approaches, including LwF [23], PASS [45], SSRE [46], FeTrIL [28], FeCAM [8], NAPA-VQ [25], FKSR [38], FCS [22] and EFC [24]. Furthermore, to verify the effectiveness of the proposed drift compensation module CADC, we conduct ablation studies comparing it with state-of-the-art drift compensation methods, including SDC [37], LDC [7], and ADC [9]. Two evaluation metrics are used to assess performance, including the Last Task Accuracy $\mathcal{A}_{last}$ and Average Incremental Accuracy $\mathcal{A}_{inc}$. Details of compared methods and evaluation metrics are provided in the Appendix.

Implementation Details. The widely used ResNet-18 [11] is adopted as the backbone and trained from scratch under the CIL setting. For all datasets, the initial task is trained

Table 4. Comparison of CADC and three state-of-the-art drift compensation methods on CIFAR-100 (10-task split) in terms of last task accuracy. Results over multiple random seeds demonstrate its strong compatibility with non-uniform semantic shifts.

Methods	Seeds											Average
	0	1	2	3	4	5	6	7	8	9	1993	
SDC	42.47	42.51	41.20	42.58	41.97	40.93	40.21	41.71	42.33	39.69	33.85	40.86
LDC	46.62	46.57	46.80	47.04	47.18	45.45	45.81	45.43	47.10	47.00	45.43	46.40
ADC	47.89	46.17	45.68	47.28	46.94	45.71	45.16	46.11	46.90	46.74	44.32	46.26
CADC	**48.44**	**47.89**	**47.88**	**48.04**	**48.38**	**47.34**	**46.93**	**47.08**	**48.51**	**48.06**	**46.44**	**47.73 (+1.33)**

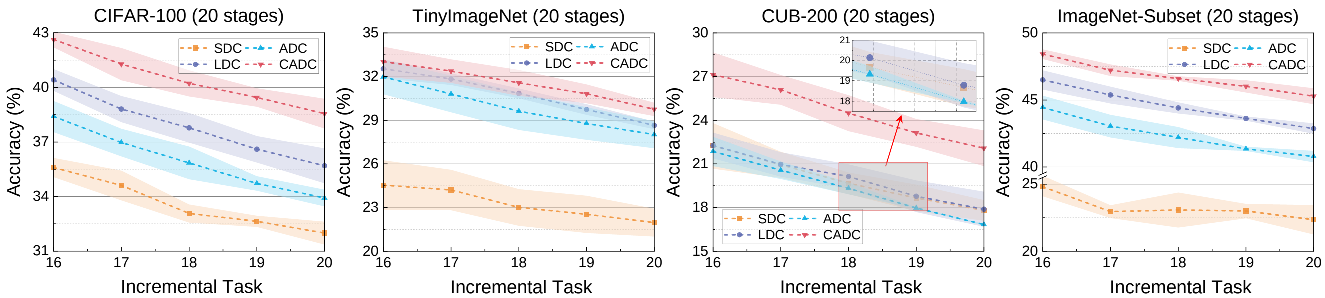

Figure 5. The drift compensation methods are evaluated across four benchmark datasets (CIFAR-100, TinyImageNet, CUB-200, and ImageNet-Subset), with performance comparisons conducted under incremental learning scenarios divided into 20 stages. The line charts specifically illustrate the performance during the last five phases. CADC is our proposed method.

Table 5. Comparison of four drift compensation methods by $\mathcal{A}_{inc}$. † indicates results from [9].

Methods	CIFAR-100		TinyImageNet	
	5 stages	10 stages	5 stages	10 stages
SDC†	64.82	58.02	50.82	40.46
LDC	70.09	59.50	51.34	46.80
ADC†	69.62	61.35	50.94	43.04
CADC	**71.16**	**62.43**	**58.23**	**51.70**

for 200 epochs with a learning rate of 0.1. For subsequent tasks, which are trained for 100 epochs, the learning rates are set to 0.05, 0.005, and 0.01 for CIFAR-100, TinyImageNet, and ImageNet-Subset, respectively. The batch size is set to 128 for CIFAR-100, and 64 for both TinyImageNet and ImageNet-Subset. For all datasets, the hyperparameters in Eq. (10) are fixed as $\gamma = 10$ and $\delta = 0.01$. The method is implemented with the PyCIL [41] framework and trained on a single NVIDIA GTX 4090 GPU. Further implementation details can be found in the Appendix.

4.2. Comparison with State-of-the-Art

Main Results. Table 1 presents a comparative analysis demonstrating the superiority of the proposed method over nine representative EFCIL approaches. Using the 10-task split of CIFAR-100 as a case study, our method achieves improvements of 1.76% in $\mathcal{A}_{last}$ and 3.87% in $\mathcal{A}_{inc}$ compared to the second-best baseline, EFC. The improvement is considerable, particularly in terms of $\mathcal{A}_{inc}$, which reflects a cumulative gain exceeding 38% across all preceding tasks. The improvements are also observed on other datasets. Fig. 4 visualizes the performance trends of various methods across different incremental tasks, demonstrating consistent superiority over the baselines at each stage.

Long Sequence Results on ImageNet-Full. We conduct experiments on the 50-task split of ImageNet-Full, where 1000 classes are evenly divided into 50 tasks. Additionally, we perform the same experiment on TinyImageNet, which contains 200 classes split into 50 tasks. According to the results in Table 2, CADC significantly outperforms the advanced LDC method on both datasets for long sequence tasks. This advantage arises because, as tasks progress, the gap between methods in prototype correction widens. Accounting for varying drift compensation degrees for classes can yield substantial improvements in long sequence tasks.

Results on Fine-Grained Datasets. To assess the effectiveness of CADC in fine-grained datasets, we compare CADC with popular approaches including SDC, LDC, and ADC. All methods are built on a model warmed up on ImageNet to ensure reasonable feature extraction capability. As shown in Table 3, CADC consistently outperforms existing methods on both fine-grained datasets, indicating that its emphasis on class-aware modeling continues to benefit drift compensation even for subtle inter-class differences.

4.3. Ablation Study

Module Ablation. We conduct a module ablation to evaluate the components in Eq. (10), using the cross-entropy

Table 6. Module ablation study under 10-task split.

Methods	CIFAR-100		TinyImageNet	
	$\mathcal{A}_{last}$	$\mathcal{A}_{inc}$	$\mathcal{A}_{last}$	$\mathcal{A}_{inc}$
Base	14.43	33.81	11.02	28.36
Base+VKD	41.80	58.62	30.73	46.25
Base+DGKD	41.34	59.10	32.88	48.09
Base+DGKD+TRSF	**47.32**	**63.18**	**38.88**	**52.54**

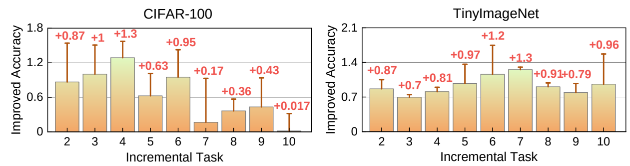

Figure 6. Accuracy improvements of DGKD over vanilla KD on incremental tasks under the 10-task split.

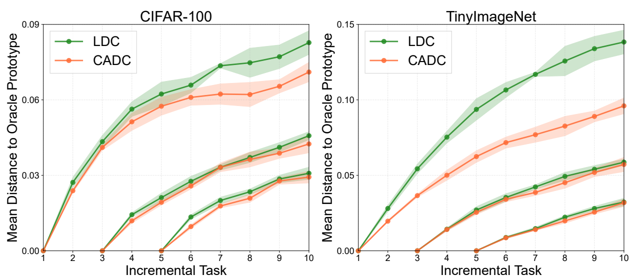

Figure 7. Trend of the average distance between corrected prototypes and oracle prototypes for tasks 1, 3, and 5.

loss $\mathcal{L}_{\mathrm{ce}}$ for new class learning as the baseline (Base). Table 6 compares vanilla KD (VKD) with our Drift-Guided KD (DGKD). DGKD outperforms VKD, with a 1.84% gain in metric $\mathcal{A}_{inc}$ on TinyImageNet. The improvement is more evident with TRSF, as drift compensation enables more accurate modeling of semantic shifts. Fig. 6 shows DGKD's improvements over VKD (with $\tau = 1$ in Eq. (6)) after each task, confirming its consistent effectiveness in CIL.

Effect of CADC. To validate the proposed CADC method, we follow [7] and use LwF [23] as the baseline to evaluate the impact of the CADC module. As shown in Table 5, CADC outperforms existing drift compensation methods, confirming the effectiveness of class-specific prototype adjustment. To simulate non-uniform semantic shifts, we conduct experiments under multiple random seeds (including the commonly used seed 1993), ensuring varied class orderings. As shown in Table 4, CADC consistently outperforms existing drift compensation methods, achieving a 1.33% improvement over the second-best method. Fig. 7 visualizes cosine distances between prototypes corrected by various drift compensation methods and oracle prototypes. Prototypes corrected by the CADC method remain closer to oracles in incremental phases, explaining its superior effectiveness. Fig. 5 shows last five task accuracy curves for drift compensation on four datasets (20-task split).

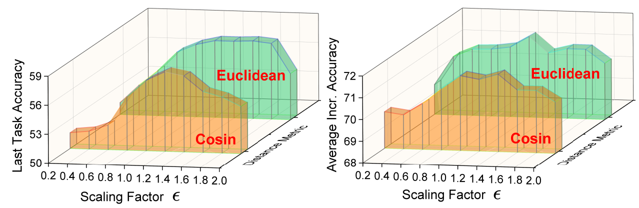

Figure 8. Analysis of scaling factor ϵ and distance metric on CIFAR-10 (5-task split).

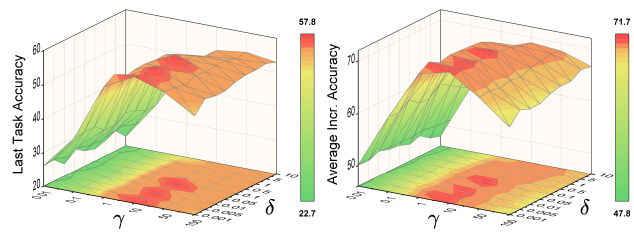

Figure 9. Analysis of γ and δ on CIFAR-10 (5-task split).

Effect of the Scaling Factor and Distance Metric. In Eq. (6), we introduce a scaling factor ϵ to control the range of compensation weights. Additionally, Eq. (3) employs Euclidean distance to model semantic shifts, although other distance metrics can also be considered. Here, we analyze the effects of these two factors on experimental outcomes. Experiments are conducted on CIFAR-10 with a 5-task split, and results are visualized in Fig. 8. The observations are as follows: (1) ϵ significantly affects both evaluation metrics $\mathcal{A}_{last}$ and $\mathcal{A}_{inc}$. As it increases from 0.2 to 2, performance rises, peaks within 1.0–1.4, and then declines. (2) Using different distance metrics to model semantic shifts yields similar trends in Fig. 8, suggesting that both Euclidean and cosine distances are effective. On this basis, we adopt Euclidean distance and set $\epsilon = 1$ for simplicity and effectiveness.

Effect of Balance Parameters. We further investigate the impact of the parameters γ and δ in Eq. (10). As shown in Fig. 9 on CIFAR-10, both parameters affect performance. Th weight γ for $\mathcal{L}_{\mathrm{dgkd}}$ yields the best results around 10, while the weight δ for $\mathcal{L}_{\mathrm{trsf}}$ should not exceed 0.1. For simplicity we set $\gamma = 10$ and $\delta = 0.01$ in all experiments.

5. Conclusions

This paper presents a class-aware drift compensation (CADC) method designed to accommodate non-uniform semantic shifts. It dynamically models class-specific semantic variations during training and applies differentiated corrections to past representations. Building on this, a drift-guided KD strategy is introduced, assigning adaptive regularization based on transfer difficulty to precisely mitigate forgetting. Extensive experiments confirm the method's effectiveness, and comparisons with existing drift compensation techniques demonstrate the superiority of CADC.

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